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HONORS CHEMISTRY • UNIT 5

Moles and Molar Quantities

Count particles by the mole, then connect particles, grams and gas volume through molar mass.

HOW TO USE THIS PACKET

Write in pencil. For every calculation show the **given with its unit, each conversion factor as a fraction so the units cancel, and the answer with units and the right number of significant figures.** Use molar masses to 0.01 g/mol, Avogadro's number 6.022×10^{23} and 22.4 L/mol for a gas at STP. Every conversion goes **through moles.** Extra practice is at the back.

What we will learn

	Topic
5.1	The mole and Avogadro's number
5.2	Molar mass
5.3	Mole conversions (particles, moles and grams)
5.4	Volume of a gas at STP
5.5	Percent composition
5.6	Empirical and molecular formulas
5.7	Hydrates

By the end of this unit I can...

- Explain what a mole is and use Avogadro's number to convert between particles and moles.
- Find the molar mass of an element or compound, including formulas with parentheses.
- Convert between particles, moles and grams using the mole map, showing every unit.
- Use 22.4 L/mol to convert between moles, grams, particles and the volume of a gas at STP.
- Calculate the percent composition by mass of a compound.
- Find an empirical formula from masses or percentages, and a molecular formula from the molar mass.
- Use the mass lost on heating to find the formula and name of a hydrate.

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5.1 The Mole and Avogadro's Number

Counting by the mole • 6.022×10^{23} • atoms, molecules and formula units

BIG IDEA

Atoms and molecules are far too small to count one at a time, so chemists count them in huge groups called **moles**. 1 mole = **6.022×10^{23}** particles, just as 1 dozen = 12. A mole is a number, not a mass.

Key vocabulary — write each definition in your own words

Term	Definition (your words)	Example / sketch
Mole	*What is it, and what does it measure?*	
Avogadro's number	*Write the value in scientific notation. What does it count?*	
Particle	*Name the three kinds of particles and say which substances each goes with.*	

HOW TO CONVERT BETWEEN PARTICLES AND MOLES

1. Decide what the particles are: atoms, molecules or formula units.
2. Write **1 mol = 6.022×10^{23} particles** as a fraction so the unit you start with cancels.
3. Moles to particles: **multiply** by 6.022×10^{23}. Particles to moles: **divide** by 6.022×10^{23}.
4. Round to the sig figs of the data and write the unit.

WORKED EXAMPLE

How many atoms are in 2.50 mol of magnesium?

Work it out here:

1. What is a "mole" in chemistry, and what is it used to measure?

2. What is the difference between an atom and a molecule?

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Convert (show the fraction setup)

	Particles and moles
3.	How many atoms are in 1.75 moles of aluminum?
4.	How many molecules are in 3.25 moles of H_2O?
5.	How many moles are in 3.4×10^{26} molecules of CO_2?
6.	How many moles are there in 3.88×10^{25} atoms of nickel?

1. Which sample contains the most particles?

A) 1 mol of iron B) 1 mol of water C) 1 mol of helium D) They all contain the same number

2. How many oxygen atoms are in 1 mol of O_2 molecules?

A) 6.022×10^{23} B) 1.204×10^{24} C) 3.011×10^{23} D) 32

5.2 Molar Mass *Grams per mole • add the atomic masses • subscripts and parentheses*

BIG IDEA

The **molar mass** of a substance is the mass of 1 mole of it, in **g/mol**. For a compound, add the atomic masses of every atom in the formula. Molar mass is the bridge between grams and moles.

HOW TO FIND A MOLAR MASS

1. List each element in the formula and **count** its atoms. A subscript outside parentheses multiplies everything inside: in $Mg(OH)_2$ there are 2 O and 2 H.
2. Multiply each count by the atomic mass (to 0.01 g/mol).
3. Add the results. Round to 0.01 g/mol and write the unit **g/mol**.

WORKED EXAMPLE

Find the molar mass of aluminum sulfate, $Al_2(SO_4)_3$.

Work it out here:

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Key vocabulary — write each definition in your own words

Term	Definition (your words)	Example / sketch
Molar mass	*What does it measure, and what unit does it have?*	
Diatomic element	*Name the seven that exist as pairs of atoms.*	

1. Define molar mass in your own words.

Calculate the molar mass of each substance (show the sum)

	Formula	Work and molar mass (g/mol)
1.	Be	
2.	N_2	
3.	O_2	
4.	F_2	
5.	Mg	
6.	Ag	
7.	LiI	
8.	$PbCO_3$	
9.	$Mg(OH)_2$	
10.	K_3PO_4	
11.	$(NH_4)_3PO_4$	
12.	$C_6H_{12}O_6$	

LAB CORNER • Weighing a mole

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A student weighs out exactly 1 mole of each of three substances: 18.02 g of water, 58.44 g of table salt (NaCl) and 55.85 g of iron.

1. (a) Which sample contains the most particles? (b) Why are the three masses different?

Answer:

2. What is the molar mass of $Ca(OH)_2$?

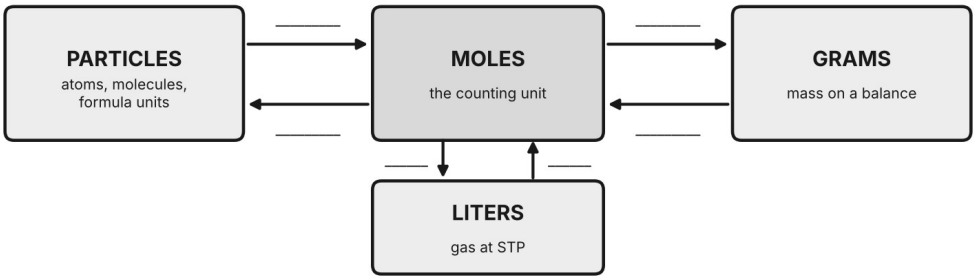
A) 57.09 g/mol B) 74.10 g/mol C) 58.10 g/mol D) 73.09 g/mol

5.3 Mole Conversions *Particles • moles • grams • the mole map*

BIG IDEA

You can never go straight from particles to grams. **Every conversion goes through moles.** The mole map shows which factor to use for each arrow: Avogadro's number for particles, molar mass for grams.

Every conversion goes through moles



The mole map. Write the conversion factor on each arrow.

STEPS

1. Write the given quantity with its unit.
2. Find the path on the mole map. Grams or particles go to **moles first**, then on to the answer.
3. Set up each factor as a fraction so the unit you want to remove cancels.
4. Calculate, round to the sig figs of the data, and label the answer with the unit and the substance.

WORKED EXAMPLE

How many molecules are in 9.00 g of water?

Work it out here:

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Solve each problem. Show givens, factors and the answer with units.

	Mole conversion problems
1.	Calculate the number of grams of CO_2 that will contain 2.5×10^{24} molecules of CO_2.
2.	How many moles of carbon tetrafluoride, CF_4, are in 176 grams of CF_4?
3.	How many moles of water are in 360.0 grams of water?
4.	How many moles of iron(III) oxide, Fe_2O_3, are there in 79.85 grams of iron(III) oxide?
5.	How many molecules of chlorine gas, Cl_2, are present in 52.5 grams of Cl_2?
6.	How many copper atoms are present in 5.6 moles of Cu_2O_3?
7.	A student measures out 2.5 moles of glucose, $C_6H_{12}O_6$. How many grams of oxygen are contained in that amount of glucose?
8.	A student measures out 15.0 grams of $CaCl_2$. How many grams of chlorine are present?

1. To convert grams of CO_2 to molecules of CO_2, the first step is to…

A) multiply by 6.022×10^{23} B) divide by the molar mass C) multiply by 22.4 D) divide by 6.022×10^{23}

2. What is the mass of 0.50 mol of H_2O?

A) 9.0 g B) 18 g C) 36 g D) 0.50 g

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5.4 Volume of a Gas at STP *22.4 L/mol • 0 °C and 1 atm • moles, grams and particles*

BIG IDEA

At **STP** (standard temperature and pressure: **0 °C and 1 atm**), 1 mole of **any** gas takes up **22.4 L**. That adds one more arrow to the mole map: moles ↔ liters of gas.

Key vocabulary — write each definition in your own words

Term	Definition (your words)	Example / sketch
STP	*What do the letters stand for, and what are the two conditions?*	
Molar volume	*How much space does 1 mol of gas fill at STP? Does the kind of gas matter?*	

STEPS

1. Moles ↔ liters: use **1 mol = 22.4 L** (STP only, gases only).
2. If the given is grams or particles, go to **moles first** with the molar mass or Avogadro's number.
3. Then use 22.4 L/mol, or go on to grams or particles for the answer.

WORKED EXAMPLE

What volume does 8.00 g of O_2 occupy at STP?

Work it out here:

What volume will each of the following occupy at STP?

	Volume at STP
1.	3.20 moles of O_2
2.	1.75 moles of NH_3
3.	37.8 moles of CO_2
4.	5.05 grams of helium
5.	60.0 grams of xenon

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Solve each problem. All gases are at STP.

	Gas problems at STP
6.	A sample of NH_3 gas occupies 75.3 L at STP. How many molecules is this?
7.	What is the mass of 575 L of ammonia (NH_3) gas at STP?
8.	A 5.00 gram sample of O_2 is in a container at STP. What is the volume of this sample?
9.	How many molecules are present in the container in the question above?
10.	How many molecules are present in 0.12 L of carbon dioxide gas at STP? How many grams?

1. Which has the greatest volume at STP?

A) 1.00 mol of N_2 B) 28.02 g of N_2 C) 6.022×10^{23} molecules of N_2 D) All three are the same

2. Why can't you use 22.4 L/mol for a gas at 25 °C?

A) 22.4 L/mol applies only at 0 °C and 1 atm B) The gas has no molar mass at 25 °C C) Moles change with temperature D) 22.4 L/mol is only for noble gases

5.5 Percent Composition *Part ÷ whole × 100 • mass percent of each element*

BIG IDEA

Percent composition tells you what part of a compound's mass comes from each element. The percents of all the elements add up to **100 %**.

HOW TO FIND PERCENT COMPOSITION

1. Find the molar mass of the compound (the **whole**).
2. For each element, find its mass in 1 mole of the compound (atomic mass × subscript): the **part**.
3. **% element = (mass of element in 1 mol ÷ molar mass of compound) × 100.**
4. Check that the percents add to 100 (within rounding).

WORKED EXAMPLE

Find the percent composition of calcium carbonate, $CaCO_3$.

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Work it out here:

Find the percent composition of every element in each compound

	Percent composition
1.	NaOH
2.	$CuBr_2$
3.	N_2S_2
4.	$(NH_4)_2S$
5.	$(NH_4)_3PO_4$

1. A 50.0 g sample of a compound is 20.0 % hydrogen by mass. How many grams of hydrogen are in the sample?

A) 2.5 g B) 10.0 g C) 20.0 g D) 40.0 g

2. Which has the greater percent oxygen by mass?

A) H_2O B) H_2O_2 C) They are the same D) Cannot be told

5.6 Empirical and Molecular Formulas

Simplest whole-number ratio • percent → grams → moles → ratio

BIG IDEA

The **empirical formula** is the simplest whole-number ratio of atoms in a compound. The **molecular formula** is the actual number of each atom, and it is a whole-number multiple of the empirical formula: **molecular = n × empirical**, where **n = molar mass ÷ empirical formula mass**.

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STEPS

1. Change each percent to grams (assume a **100 g** sample), or use the grams given.
2. Change each mass to **moles** (÷ atomic mass).
3. Divide every mole value by the **smallest**. These are the subscripts.
4. If a result is near x.5, multiply **all** by 2 (near x.33 or x.67, by 3). Do not round 1.5 up to 2.
5. Molecular formula: n = molar mass ÷ empirical mass; multiply every subscript by n.

WORKED EXAMPLE

A compound is 40.0 % C, 6.7 % H and 53.3 % O and has a molar mass of 180 g/mol. Find its empirical and molecular formulas.

Work it out here:

Solve each problem

	Empirical and molecular formula problems
1.	Determine the empirical formula of the compound if it contains 87.3 g Na, 121.5 g S and 91.2 g O.
2.	A compound contains 85.65 % Bi, 6.56 % O and 7.79 % F. What is the empirical formula of this compound?
3.	Propene has an empirical formula of CH_2. What is its molecular formula if it has a molar mass of 42.09 g/mol?
4.	The percent composition of an unknown substance is 75.42 % carbon, 6.63 % hydrogen, 8.38 % nitrogen and 9.57 % oxygen. If its molar mass is 334.0 g/mol, what is its empirical and molecular formula?

1. What is the empirical formula of $C_6H_{12}O_6$?

A) CHO B) CH_2O C) $C_2H_4O_2$ D) $C_6H_{12}O_6$

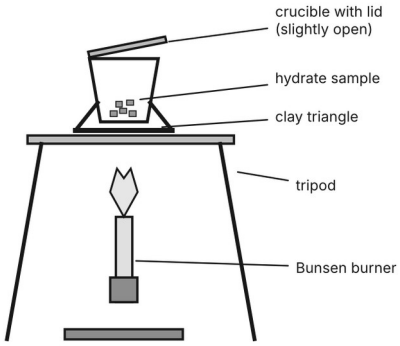
2. A compound has the empirical formula CH and a molar mass of 78 g/mol. Its molecular formula is…

A) CH B) C_2H_2 C) C_6H_6 D) C_6H_{12}

5.7 Hydrates *Water in the crystal • heating • the formula and the name*

BIG IDEA

A **hydrate** is an ionic compound with a fixed number of water molecules built into its crystal, such as $CuSO_4 \cdot 5H_2O$. Heating drives the water off and leaves the **anhydrous** ("without water") salt. The mass lost is the mass of water, so you can compare moles of water to moles of salt.



Heating a hydrate in a crucible. Mass the crucible, the hydrate and the anhydrous salt.

1	2	3	4	5	6	7	8	9	10
mono-	di-	tri-	tetra-	penta-	hexa-	hepta-	octa-	nona-	deca-

HOW TO FIND THE FORMULA OF A HYDRATE

1. mass of water = mass of hydrate − mass of anhydrous salt.
2. Change both to **moles**: water ÷ 18.02 g/mol, salt ÷ its molar mass.
3. x = moles of H_2O ÷ moles of salt (divide by the smaller). Round to the **nearest whole number**.
4. Write the formula **salt · xH_2O** and the name **salt name + prefix + "hydrate"**.

WORKED EXAMPLE

A 7.36 g sample of a hydrate of calcium chloride is heated until no water remains. The $CaCl_2$ left has a mass of 5.55 g. Find the formula and name of the hydrate.

Work it out here:

Find the formula and the name of each hydrate

	Hydrate problems
1.	A hydrate of magnesium sulfate has a mass of 13.52 g. This sample is heated until no water remains. The $MgSO_4$ anhydrate has a mass of 6.60 g. Find the formula and name of the hydrate.

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	Hydrate problems
2.	A sample of copper(II) sulfate hydrate has a mass of 3.97 g. After heating, the $CuSO_4$ that remains has a mass of 2.54 g. Determine the correct formula and name of the hydrate.
3.	A sample of the hydrate of sodium carbonate has a mass of 8.85 g. It loses 1.28 g when heated. Find the formula and the name of the hydrate.
4.	A 16.4 g sample of hydrated calcium sulfate is heated until all the water is driven off. The calcium sulfate that remains has a mass of 13.0 g. Find the formula and the chemical name of the hydrate.
5.	A hydrate is determined to be 45.43 % water and 54.57 % $CoCl_2$. Find the chemical formula and name for this hydrate. (Hint: assume that there is 100 g total of hydrate compound.)

1. After a hydrate is heated, its mass decreases. The mass lost is the mass of…

A) the salt B) the water C) the crucible D) air

2. Which is the correct name for $CoCl_2 \cdot 6H_2O$?

A) cobalt chloride hexahydrate B) cobalt(II) chloride hexahydrate C) cobalt(II) chloride sixhydrate D) cobalt(II) chlorate hexahydrate

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PRACTICE

Extra Practice — Unit 5

Show your work. Every conversion goes through moles.

A. Which conversion? (give the factor you would use)

	Situation	Factor
a.	Grams of a compound to moles of the compound.	
b.	Moles of molecules to the number of molecules.	
c.	Liters of a gas at STP to moles of the gas.	
d.	Percent composition to grams (to start an empirical formula).	

B. Calculate

	Problems
a.	How many molecules are in 8.50 g of NH_3?
b.	What is the percent by mass of carbon in glucose, $C_6H_{12}O_6$?
c.	A compound is 52.14 % C, 13.13 % H and 34.73 % O and has a molar mass of 46.07 g/mol. Find its molecular formula.
d.	What is the mass of 5.00 L of CO_2 at STP?

C. Lab data: a hydrate of barium chloride

A student heats a hydrate of barium chloride in a crucible. Complete the table, then find x in $BaCl_2 \cdot xH_2O$ (molar mass of $BaCl_2$ = 208.23 g/mol).

Measurement	Data	Your calculation
Mass of empty crucible	18.42 g	

Measurement	Data	Your calculation
Mass of crucible + hydrate	21.67 g	
Mass of crucible + anhydrous $BaCl_2$ after heating	21.19 g	
Mass of the hydrate		
Mass of the anhydrous $BaCl_2$		
Mass of water lost		

1. Find the moles of water, the moles of $BaCl_2$, the value of x, and write the formula and name.

Answer:

D. Think

1. One mole of copper (63.55 g) and one mole of helium (4.00 g) have different masses. Explain why they still contain the same number of atoms.

2. A student finds the formula of a hydrate to be $CuSO_4 \cdot 4.2H_2O$. Give one possible reason the ratio is not a whole number, and say what the student should report.

3. Which of these is NOT 1 mole of the substance?

A) 6.022×10^{23} atoms of Na B) 22.99 g of Na C) 22.4 L of Na at STP D) 58.44 g of NaCl

4. A compound has the empirical formula CH_2O and a molar mass of 60.06 g/mol. Its molecular formula is…

A) CH_2O B) $C_2H_4O_2$ C) $C_3H_6O_3$ D) $C_6H_{12}O_6$